

Homework — 1.2.4 Types of Programming Language

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.2.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Write the letter of the correct definition in the **Match** column next to each term. Each letter is used exactly once. [3] (AO1)

Term	Match	Letter	Definition
Programming paradigm		A	A subclass acquires the attributes and methods of a superclass, and may add its own or override them
Procedural language		B	A style or approach to writing programs, e.g. procedural, object-oriented, assembly
Assembly language		C	Keeping an object's attributes private so they can only be read or changed through its public methods
Class		D	A low-level language using mnemonics, where each instruction translates to one machine-code instruction
Inheritance		E	A template that defines the attributes and methods from which objects are created
Encapsulation		F	A language built from sequence, selection, iteration and subroutines, setting out step by step <i>how</i> a task is done

2. The following Little Man Computer (LMC) program inputs two numbers.

```
      INP
      STA first
      INP
      SUB first
      BRP big
      LDA first
      OUT
      HLT
big    ADD first
      OUT
      HLT
first  DAT
```

(a) Complete the trace table for the inputs **9** then **5**. Exactly **eight** instructions are executed, and the first row has been done for you. In each row, show the values **after** the instruction finishes (copy a value down if it has not changed; only write in **Output** when a value is output). [3] (AO2)

Instruction executed	ACC	first	Output
INP	9		

(b) State the purpose of the program for **any** two inputs. [1] (AO2)

3. A processor executes the instruction `LDA 4`. The relevant contents of memory and registers are:

Location	Contents
Address 4	7
Address 6	9
Address 7	2
Index register	2

Complete the table to show the value loaded into the Accumulator when the operand `4` is interpreted using each addressing mode. [4] (AO2)

Addressing mode	Value loaded into ACC
Immediate	
Direct	
Indirect	
Indexed	

4. A studio is writing a game containing many types of character (wizards, archers, guards) that all share common behaviour such as moving and holding items. Explain why an **object-oriented** paradigm is well suited to this program. Refer to **inheritance** and **encapsulation** in your answer. [3] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

5. (1.2.3) Write **True** or **False** for each statement about software development methodologies. If a statement is false, write a correction in the final column. [3] (AO1)

Statement	True / False	Correction if false
In the Waterfall model, the stages are completed in sequence, with each stage finished before the next begins.		
In the Waterfall model, the client normally sees working software early and often during development.		
Extreme programming is an example of the Waterfall model.		

6. (1.2.2) Explain the difference between **static** and **dynamic** linking of library code. [2] (AO1)

7. (1.1.1) State the purpose of the **Accumulator (ACC)** register. [1] (AO1)
